

# Training: micro\_tcn

The separable causal TCN, 16 filters, and its four variants.

micro\_tcn was trained once from scratch in round 1 (about 11 hours, 45 full passes). Its variants either change one setting from scratch (fewer filters, more inputs) or continue from its weights.

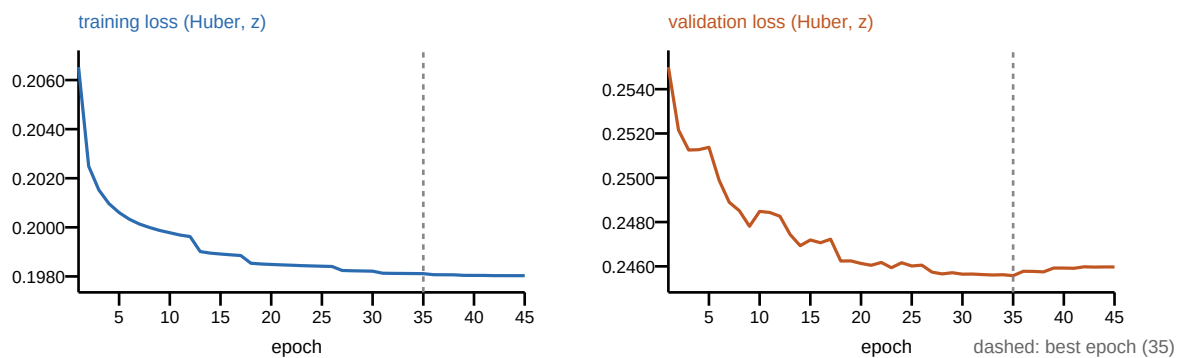
## How this family is trained

- Inputs: the (30, 2) z-scored RR/dRR sequence + the 9 'seq' features; residual target; the shared Keras recipe. BatchNorm layers train normally and are folded into the convolutions at int8 export.
- Variants compare against their parent with the same seed (211), one change each.

## Runs

### micro\_tcn - round 1 base

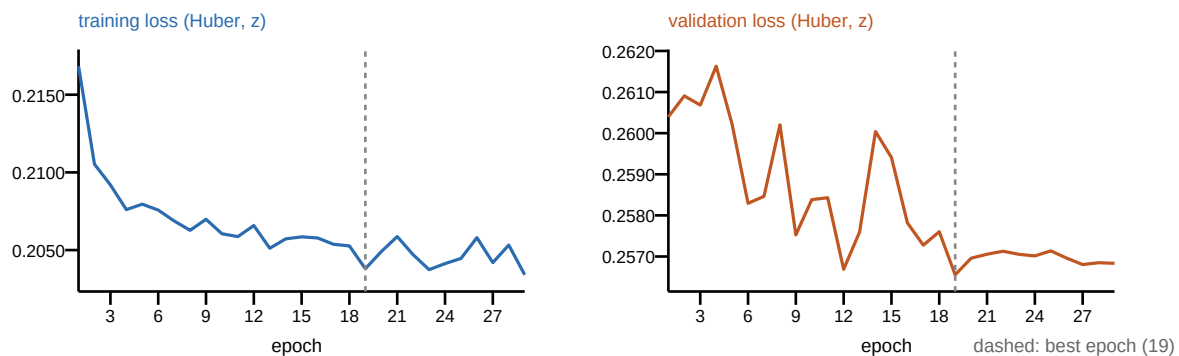
| item                    | value                                                                                                                                                                                                                                |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| command                 | python src/train.py --models micro_tcn --tag micro_tcn                                                                                                                                                                               |
| start                   | from scratch (random initialisation, seed 211)                                                                                                                                                                                       |
| inputs                  | 30-beat RR sequence (z-RR, z-dRR) + 9 tabular features (set seq)                                                                                                                                                                     |
| target                  | z-scored correction over the RLS forecast (dRR - rls_pred)                                                                                                                                                                           |
| epoch                   | 20,755,456 windows = one full pass over the 20,755,925 training windows (round-1 train.py)                                                                                                                                           |
| budget                  | 45 of at most 100 epochs (early stopping, patience 10); 933,995,520 windows drawn = 45.0 passes over the training set                                                                                                                |
| learning rate           | 0.001 at the start, halved 7 times on validation plateaus, 1e-05 at the end                                                                                                                                                          |
| best epoch              | 35 (validation loss 0.2550 after epoch 1, 0.2456 at best); its weights are the saved model                                                                                                                                           |
| time                    | 10.86 h on one GPU                                                                                                                                                                                                                   |
| int8                    | re-exported 2026-10-08 with best-of-K calibration: chose draw n2000_s2 (2000 windows), int8-vs-float gap +0.31% on the held-out training windows (plain 20k min/max: +1.06%); full validation: int8 33.47 vs float 32.93 ms (+1.63%) |
| validation RMSE (float) | 32.93 ms on all 1,421,225 validation windows (-11.3% vs rls_ar 37.14)                                                                                                                                                                |



- The int8 file was re-exported twice: with 20,000 windows (old file kept as model\_int8\_prev\_calib.tflite) and then best-of-K (20k file kept as model\_int8\_20k.tflite).

### micro\_tcn\_f8 - round 2, 8 filters, from scratch

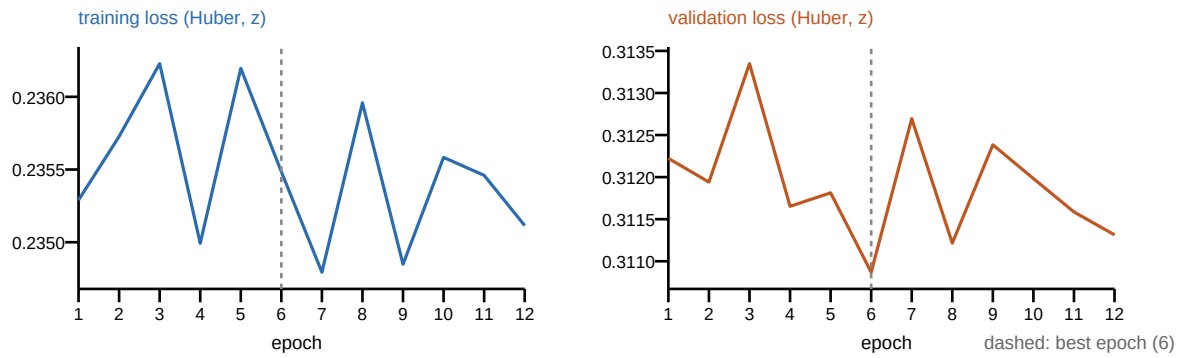
| item                    | value                                                                                                                                                                                                                             |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| command                 | python src/train.py --models micro_tcn_f8 --tag micro_tcn_f8                                                                                                                                                                      |
| start                   | from scratch (random initialisation, seed 211)                                                                                                                                                                                    |
| inputs                  | 30-beat RR sequence (z-RR, z-dRR) + 9 tabular features (set seq)                                                                                                                                                                  |
| target                  | z-scored correction over the RLS forecast (dRR - rls_pred)                                                                                                                                                                        |
| epoch                   | 1,999,872 windows drawn at random with replacement (512 x 3906 steps)                                                                                                                                                             |
| budget                  | 29 of at most 100 epochs (early stopping, patience 10); 57,996,288 windows drawn = 2.8 passes over the training set                                                                                                               |
| learning rate           | 0.001 at the start, halved 6 times on validation plateaus, 1.6e-05 at the end                                                                                                                                                     |
| best epoch              | 19 (validation loss 0.2604 after epoch 1, 0.2566 at best); its weights are the saved model                                                                                                                                        |
| time                    | 0.59 h on one GPU, 18.6 ms per step                                                                                                                                                                                               |
| int8                    | re-exported 2026-10-07 after the 500-window calibration bug, min/max calibration on 20,000 random training windows; first 20k validation windows: 35.13 vs 34.70 ms float; full validation: int8 34.95 vs float 34.73 ms (+0.63%) |
| validation RMSE (float) | 34.73 ms on all 1,421,225 validation windows (-6.5% vs rls_ar 37.14)                                                                                                                                                              |



- Re-exported with 20,000 windows after the calibration bug.

### micro\_tcn\_dad - round 3, fine-tune on closed-loop data

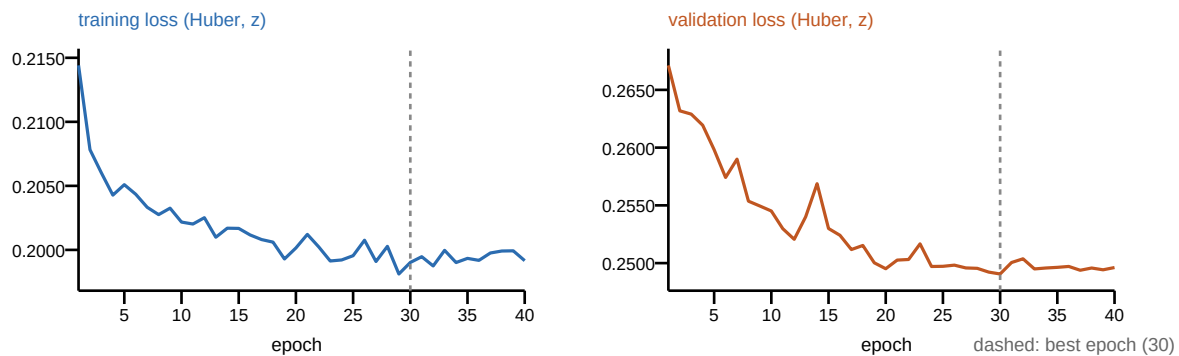
| item                    | value                                                                                                                                                                                                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| command                 | python src/train.py --models micro_tcn --init_from runs/micro_tcn --dad_filler runs/micro_tcn --dad_share 0.5 --lr 0.0003 --epochs 30 --patience 6 --tag micro_tcn_dad                                                                                                                    |
| start                   | fine-tune of micro_tcn: its weights, features, residual flag and normalisation; a fresh AdamW optimiser                                                                                                                                                                                   |
| inputs                  | 30-beat RR sequence (z-RR, z-dRR) + 9 tabular features (set seq)                                                                                                                                                                                                                          |
| target                  | z-scored correction over the RLS forecast (dRR - rls_pred)                                                                                                                                                                                                                                |
| epoch                   | 1,999,872 windows drawn at random with replacement (512 x 3906 steps)                                                                                                                                                                                                                     |
| budget                  | 12 of at most 30 epochs (early stopping, patience 6); 23,998,464 windows drawn = 1.2 passes over the training set                                                                                                                                                                         |
| learning rate           | 0.0003 at the start, halved 2 times on validation plateaus, 7.5e-05 at the end                                                                                                                                                                                                            |
| best epoch              | 6 (validation loss 0.3122 after epoch 1, 0.3109 at best); its weights are the saved model                                                                                                                                                                                                 |
| time                    | 0.29 h on one GPU, 21.6 ms per step                                                                                                                                                                                                                                                       |
| closed-loop data        | micro_tcn filled 4,220 training segments of 5000 beats (20,511,092 windows, 10%-80% dropped per segment, seed 0) in 244 s on CPU; each training window is closed-loop with probability 0.5 (realised 0.503); early stopping on a 50/50 mix of ordinary and closed-loop validation windows |
| int8                    | re-exported 2026-10-08 with best-of-K calibration: chose draw n2000_s1 (2000 windows), int8-vs-float gap +0.31% on the held-out training windows (plain 20k min/max: +0.37%); full validation: int8 33.71 vs float 33.47 ms (+0.73%)                                                      |
| validation RMSE (float) | 33.47 ms on all 1,421,225 validation windows (-9.9% vs rls_ar 37.14); parent 32.93 ms; closed-loop validation windows 42.43 ms (parent 42.39, rls_ar 44.28)                                                                                                                               |



- Its first attempt was stopped by the coordinator because of the tf.data index bug and deleted; this is the re-queued run.
- The validation loss is higher than in the other runs because it is measured on a 50/50 mix with closed-loop windows, which are harder.

#### ***micro\_tcn\_rsa - round 4, + gated RSA features, from scratch***

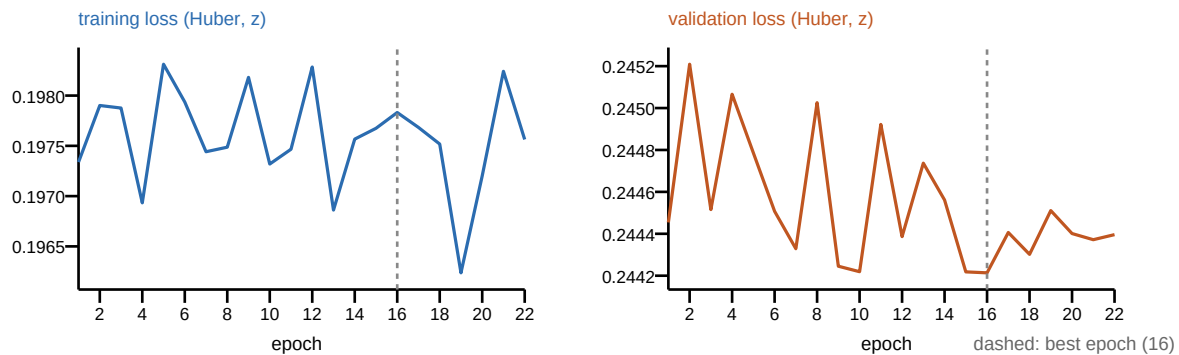
| item                    | value                                                                                                                                                                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| command                 | python src/train.py --models micro_tcn --features seq_rsa --tag micro_tcn_rsa                                                                                                                            |
| start                   | from scratch (random initialisation, seed 211)                                                                                                                                                           |
| inputs                  | 30-beat RR sequence (z-RR, z-dRR) + 13 tabular features (set seq_rsa)                                                                                                                                    |
| target                  | z-scored correction over the RLS forecast (dRR - rls_pred)                                                                                                                                               |
| epoch                   | 1,999,872 windows drawn at random with replacement (512 x 3906 steps)                                                                                                                                    |
| budget                  | 40 of at most 100 epochs (early stopping, patience 10); 79,994,880 windows drawn = 3.9 passes over the training set                                                                                      |
| learning rate           | 0.001 at the start, halved 6 times on validation plateaus, 1.6e-05 at the end                                                                                                                            |
| best epoch              | 30 (validation loss 0.2671 after epoch 1, 0.2491 at best); its weights are the saved model                                                                                                               |
| time                    | 0.77 h on one GPU, 17.7 ms per step                                                                                                                                                                      |
| int8                    | best-of-K calibration: chose draw n2000_s2 (2000 windows), int8-vs-float gap +0.05% on the held-out training windows (plain 20k min/max: +0.32%); full validation: int8 34.03 vs float 33.86 ms (+0.49%) |
| validation RMSE (float) | 33.86 ms on all 1,421,225 validation windows (-8.8% vs rls_ar 37.14)                                                                                                                                     |



- A new input layer cannot load the parent's weights, so it started from scratch with the 2M-window epochs: about 3.9 passes vs about 45 for micro\_tcn. The comparison is confounded by this budget.

## micro\_tcn\_rsa\_ws - round 4, + gated RSA features, warm start

| item                    | value                                                                                                                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| command                 | python src/train.py --models micro_tcn --features seq_rsa --init_from runs/micro_tcn --lr 0.0003 --epochs 30 --patience 6 --tag micro_tcn_rsa_ws                                                                                              |
| start                   | warm start from micro_tcn: every weight copied, the 4 new inputs (rsa_f, rsa_amp, rsa_q, rsa_drr) start at zero weight, so at step 0 it computes exactly the parent's function (checked: max difference 9.5e-07 z on real validation windows) |
| inputs                  | 30-beat RR sequence (z-RR, z-dRR) + 13 tabular features (set seq_rsa)                                                                                                                                                                         |
| target                  | z-scored correction over the RLS forecast (dRR - rls_pred)                                                                                                                                                                                    |
| epoch                   | 1,999,872 windows drawn at random with replacement (512 x 3906 steps)                                                                                                                                                                         |
| budget                  | 22 of at most 30 epochs (early stopping, patience 6); 43,997,184 windows drawn = 2.1 passes over the training set                                                                                                                             |
| learning rate           | 0.0003 at the start, halved 5 times on validation plateaus, 1e-05 at the end                                                                                                                                                                  |
| best epoch              | 16 (validation loss 0.2445 after epoch 1, 0.2442 at best); its weights are the saved model                                                                                                                                                    |
| time                    | 0.45 h on one GPU, 18.6 ms per step                                                                                                                                                                                                           |
| int8                    | best-of-K calibration: chose draw n2000_s2 (2000 windows), int8-vs-float gap +0.33% on the held-out training windows (plain 20k min/max: +0.92%); full validation: int8 33.47 vs float 32.93 ms (+1.63%)                                      |
| validation RMSE (float) | 32.93 ms on all 1,421,225 validation windows (-11.3% vs rls_ar 37.14); parent 32.93 ms                                                                                                                                                        |



- The clean test of the RSA inputs: same function as micro\_tcn at step 0, then the fine-tune schedule.

## What the training showed

- micro\_tcn's round-1 run reached the 1e-5 learning-rate floor with a flat validation loss: it is trained to convergence.
- Halving the filters (f8) costs about 1.8 ms; DAD and the warm-started RSA inputs move the validation RMSE by less than 0.6 ms. The selection in the closed loop went to the larger tcn\_lite\_ft.

## Abbreviations and acronyms used in this document

| term | meaning                                                                                                                                    |
|------|--------------------------------------------------------------------------------------------------------------------------------------------|
| DAD  | data as demonstrator: training on inputs that contain the model's own closed-loop fills, with the true target (a remedy for exposure bias) |
| dRR  | beat-to-beat change of the RR interval: $dRR[n+1] = RR[n+1] - RR[n]$ (ms); the quantity predicted                                          |
| _ft  | run-name suffix: fine-tuned (training continued from the parent's weights)                                                                 |
| HRV  | heart rate variability                                                                                                                     |
| int8 | 8-bit integer: weights and activations quantised to 8-bit integers, as run on the microcontroller; _int8 = the int8 version of a run       |
| ms   | milliseconds                                                                                                                               |
| RMSE | root mean square error (ms): the main accuracy measure, lower is better                                                                    |
| RR   | RR interval: time between two consecutive heartbeats (R waves of the ECG), in ms                                                           |
| RSA  | respiratory sinus arrhythmia: the oscillation of the heart rate driven by breathing                                                        |
| seq  | feature-set / model-kind name: sequence model (takes the raw RR window plus tabular features)                                              |
| TCN  | temporal convolutional network (causal, dilated 1-D convolutions)                                                                          |
| tcn  | temporal convolutional network (run-name part)                                                                                             |
| z    | z-score: $(value - mean) / standard\ deviation$ , with statistics from the training subjects                                               |